

- 5 (a) (i) The magnetic flux density of a magnetic field is defined as the force per unit current per unit length (of conductor) acting on a straight current-carrying conductor placed at right angles to the magnetic field [B1].
- (ii) Current in clockwise direction through wire [A1]
- (iii)
$$B = \mu_0 n I = 4\pi \times 10^{-7} \times \frac{50}{0.30} \times 2.0 = 4189 \times 10^{-4} \text{ T [A1]}$$
- (iv)
$$F = Bqv$$

$$= (4.189 \times 10^{-4})(1.6 \times 10^{-19})(10) \text{ [C1]}$$

$$= 6.7 \times 10^{-22} \text{ N [A1]}$$
- (v) The magnetic force is acting perpendicular to the direction of travel/ velocity of the electron [M1] and this will only affect direction of velocity and not its magnitude [A1]

Alternative answer:

The magnetic force is acting perpendicular to travel/ velocity of the electron [M1] and therefore no (net) work done [A1] on the electron, resulting in no change in kinetic energy

- (b) (i) No net force acting on electron,
Magnetic force = Electric force [C1]
 $qE = Bqv$
 $E = Bv = 4.19 \times 10^{-3} \text{ Vm}^{-1} \text{ [A1]}$
- (ii) Downwards vertical arrow labelled E [B1]